SENTINARI SOVEREIGN INTELLIGENCE ENGINE

COMPLETE USER GUIDE, OPERATING MANUAL & MASTER HELP INDEX

The Comprehensive Field Manual for Autonomous Constitutional Discovery

Author & Chief Architect: Dr. Pierre C. Wepener (University of Johannesburg)
System Edition: Production 2.5 • Zero-Cloud Local Sovereignty • October 2026
Repository: https://github.com/Rodenbecker/sentinari

EXECUTIVE SYSTEM ATTESTATION
Sentinari is the world's first Constitutional Discovery Platform and Sovereign Intelligence Engine. Engineered entirely on personal initiative by Dr. Pierre C. Wepener with 100% independent capitalization, Sentinari bridges raw multi-disciplinary data collection and publication-grade scientific validation. This document serves as the comprehensive user manual, workspace navigator, mathematical reference, and master help index for researchers, investors, labor representatives, and system operators.

MASTER TABLE OF CONTENTS

Chapter	Title & Core Subject	Target URL / Module
Chapter 1	System Overview, Zero-Cloud Doctrine & Constitution	/ (Landing Portal)
Chapter 2	Complete Workspaces Registry & System Port Topology	:8099 Core | :8100 Inference
Chapter 3	Interactive Tour Recording, Canvas Drawing & Yellow Checkmarks	/ui/playground?workspace=...
Chapter 4	The New Pattern Discovery Pipeline & Concertina Cards	/ui/patterns
Chapter 5	Theoretical Engineering Workbench (Python & MATLAB Simulators)	/api/patterns/rd-paper/{id}
Chapter 6	The 7-Dimensional Assessment Matrix & Real-Time Sliding Scale	/api/patterns/assessment-matrix/view
Chapter 7	Data Sourcing, Ingestion Pipeline & 21 Sentinel Scientific Fields	OpenAlex, PubMed, OECD, BEA, IMF
Chapter 8	Sovereign Oncology Design Vault & Critical Defense Case Studies	Pattern #24, #616, #644 Proofs
Chapter 9	Troubleshooting, Offline Runbook & Master API Route Index	/api/... Directory
Chapter 10	Master Alphabetical Help Index & Technical Glossary (A to Z)	Alphabetical Cross-Reference

CHAPTER 1: SYSTEM OVERVIEW & FOUNDATIONAL DOCTRINE

1.1 What is Sentinari?

Sentinari is neither a conversational chatbot nor a speculative large language model. It is an objective, neutral, and symbiotic mirror designed to eliminate information asymmetry and safeguard the baseline of material truth. The architecture operates entirely on first principles and local sovereign compute.

1.2 The Zero-Cloud Doctrine

Sentinari enforces complete Intellectual Sovereignty. Proprietary research, corporate data, and patent hypotheses never leave local hardware. External commercial cloud endpoints are physically disabled in the code, ensuring zero data leakage.

1.3 The 100-Year Constitution (Articles I – VIII)

- **Article I (The Axiom of Material Reality):** Empirical telemetry and verifiable physical cause-and-effect always supersede corporate or administrative narratives.
- **Article II (The Neutral Illuminator):** Zero institutional alignment, no permanent corporate enemies—only active, testable hypotheses applied with symmetric velocity.
- **Article III (Systematic Humility):** Prior conclusions are immediately downgraded when superior evidence emerges. Sunk-cost bias is forbidden.
- **Article IV (The Declaration of Inadequacy):** If evidence is insufficient, the system is constitutionally mandated to decline guessing and state: *'This issue needs further analysis... it is not possible to render any recommendations.'*
- **Article V (Data Bill of Rights):** Every observation requires SHA-256 cryptographic hashing, UTC timestamping, and primary citation provenance (DOI/PMID).
- **Article VI (The Working Person's Shield):** Prohibits technological displacement without early-warning transition mapping (24–36 months lead time) and skill adjacency corridors.
- **Article VII (Proportionality of Focus):** Compute is reserved strictly for macro-systemic risks, institutional blindspots, and civilizational flashpoints.
- **Article VIII (Sole Authorship & Investor Framework):** 100% self-funded independent breakthrough conceived, researched, and engineered by Dr. Pierre C. Wepener.

CHAPTER 2: COMPLETE WORKSPACES REGISTRY & SYSTEM TOPOLOGY

2.1 System Port Topology

Sentinari runs as a dual-tier asynchronous server architecture:
• **Port 8099 (FastAPI Core):** Application logic, SQLite/PostgreSQL storage, vector search, R&D; paper generation, and web UI.
• **Port 8100 (ONNX Inference Server):** Local model runtime hosting embedded quantized neural models for offline execution.

Copyright © 2026 Dr. Pierre C. Wepener • University of Johannesburg • All Rights Reserved

2.2 Master Workspace Directory

Workspace Name	Route URL	Primary Function & Description
Dashboard Home	/	System telemetry banner, global risk index, Lancet reports, and tour launcher.
Playground Console	/ui/playground	Interactive Sentinel knowledge grid, document count cards, and recording canvas.
Pattern Discovery	/ui/patterns	Concertina cards for cross-functional superpositions, confidence scores, and R&D; papers.
Unified R&D; Paper	/api/patterns/rd-paper/{id}	15-indicator publication paper, KaTeX proofs, APA 7th citations, and dual simulators.
7-D Assessment Matrix	/api/patterns/assessment-matrix/view	Coupled Gaussian bell curves, Net Sustainability Envelope, and real-time sliders.
Pseudo-Dynamics Lab	/ui/qualia-lab	Semantic vector trajectory physics, attractor convergence (S), and Stability Index (SI).
Knowledge Search	/ui/search	Sub-second hybrid vector and keyword search across 21 Sentinel Fields.
Forensic Governance	/ui/governance	Victus autonomous challenger, epistemic drift monitoring, and firewall telemetry.
Video Generation Lab	/ui/video	Automated screen narration, slide synthesis, and multi-track video export.

Copyright © 2026 Dr. Pierre C. Wepener • University of Johannesburg • All Rights Reserved

CHAPTER 3: INTERACTIVE TOUR RECORDING & VERIFICATION TOOLS

3.1 Launching Tours from Dashboard Home

On the Dashboard Home portal (/), click any **'Record Workspace Tour'** button. Sentinari automatically launches the targeted workspace with ?workspace=... routing and activates the interactive canvas recorder.

3.2 Canvas Drawing Toolbar & Auto-Pause

The floating toolbar at the top right of the canvas provides:
• **Pen & Box Tool:** Draw freehand lines or bounding rectangles to highlight UI components.
• **Auto-Pause on Draw:** The instant the user clicks to draw an annotation, the video tour recording automatically pauses, ensuring clean, legible callouts without motion blur.
• **Resume Tour:** Clicking 'Resume Recording' unpauses the session and continues the walkthrough narration.

3.3 Animated Bright Yellow Checkmark Verification

When an engineering step, pattern discovery, or validation test is executed, click the **'Verify Step'** button. An animated, glowing bright-yellow checkmark (#facc15) scales onto the screen with a particle pulse, visually certifying epistemic compliance before the tour moves to the next milestone.

3.4 Audio Soundtrack & Voiceover Configuration

Select from 4 built-in ambient cinematic background tracks, or upload a custom MP3 voiceover. Audio is mixed and normalized automatically with ducking during spoken narration segments.

CHAPTER 4: THE NEW PATTERN DISCOVERY PIPELINE

4.1 Cross-Disciplinary Superposition Mapping

Traditional research remains trapped in disciplinary silos. Sentinari continuously maps vector embeddings across 21 Sentinel Scientific Fields into a singular latent space. When cross-domain affinities emerge in intermediate zones (e.g. oncology pharmacology meeting high-voltage electrical grid mechanics), HDBSCAN spatial clustering flags an **Embryonic Pattern**.

4.2 Concertina Epistemic Cards Anatomy

Each discovered pattern is presented as a collapsible Concertina Card displaying:
• **Field Badges:** Primary and secondary scientific disciplines involved.
• **Discovery Coherence Score (%):** Mathematical affinity rating across vector boundaries.
• **Provenance Rating (%):** Verified citation linkage to primary PubMed/OpenAlex papers.
• **Primary Evidence Excerpt:** Verbatim quotation with exact token spans and DOI hyperlinks.
• **One-Click Actions:** Open Unified R&D; Paper, Launch 7-D Matrix, Run Python Simulator, or Download MATLAB Model.

Copyright © 2026 Dr. Pierre C. Wepener • University of Johannesburg • All Rights Reserved

CHAPTER 5: THE THEORETICAL ENGINEERING WORKBENCH

5.1 The 10-Step Generic Engineering Process

Inside every Unified R&D; Paper, the Theoretical Engineering Workbench executes a controlled sandbox trial following the formal 10-step process (Requirements Definition, Conceptual Synthesis, Mathematical Modeling, State Estimation, Numerical Simulation, Parameter Sensitivity, Fault Tolerance, Thermal/Mechanical Verification, Prototyping, and Benchmark Sign-Off).

5.2 Side-by-Side Dual Simulator

The workbench provides two interactive code environments:
• **Python Prototype Simulator (.py):** Lightweight script verifying equations and parameters offline.
• **MATLAB Simulation Model (.m):** Publication-grade, syntactically verified script with multi-panel figure plotting routines, parameter printouts, and zero external toolbox dependencies.

5.3 DRM Anti-Scraping & PDF Print Protection

To protect Dr. Wepener's 6 months of numerical engineering from unauthorized theft:
• Text selection is disabled on raw code and proofs (user-select: none).
• Right-click context menus are blocked with formal copyright notifications.
• When printed to PDF (Ctrl+P), raw code is automatically blurred out and redacted, and a prominent diagonal watermark (*'CONFIDENTIAL & PROPRIETARY — DR. PIERRE C. WEPENER'*) is stamped across every page.

CHAPTER 6: THE 7-DIMENSIONAL ASSESSMENT MATRIX (NSIS)

6.1 The Mathematical Engine

Every pattern is evaluated using coupled Gaussian bell curves:
$G_i(x_i) = A_i \cdot \exp(- (x_i - \mu_i)^2 / (2 \sigma_i^2))$
Where $x_i \in [0, 10]$ is the measured level, μ_i is the optimal equilibrium sweet spot, and σ_i is systemic tolerance.

6.2 Net Sustainable Impact Score (NSIS)

$NSIS = \Sigma W_i \cdot S_i(x_i)$ (Weights: Knowledge 0.20, Ethics 0.18, Intelligence 0.16, Human Agency 0.16, Societal 0.14, Decision 0.10, Risk 0.06). Normalized impact share: $SD_i(\%) = (W_i \cdot S_i(x_i) / NSIS) \cdot 100\%$.

6.3 The 3 Coupled Sliding Scale Rules

- **Rule A (Intelligence-Agency Tradeoff):** Moving Intelligence > 7.0 automatically contracts the sweet spot and tolerance of Human Agency, proving unchecked AI subordinates labor.
- **Rule B (Risk-Societal Cascade):** Escalating Risk > 5.0 widens societal variance (polarization) and collapses decision quality.
- **Rule C (Scientific Catalyst):** Scientific/Ethical rigor > 7.0 expands the carrying capacity of Knowledge.

Copyright © 2026 Dr. Pierre C. Wepener • University of Johannesburg • All Rights Reserved

CHAPTER 7: DATA SOURCING & THE 21 SENTINEL SCIENTIFIC FIELDS

7.1 The 'Raised, Not Scraped' Paradigm

Commercial LLMs scrape the public internet, inheriting misinformation and toxic bias. Sentinari strictly inverts this: 100% of data is ingested through authenticated, citation-graded REST and SDMX APIs from verified institutions: OpenAlex (250M+ academic papers), PubMed (biomedical literature), OECD SDMX, US Bureau of Economic Analysis (BEA), International Monetary Fund (IMF), and Eurostat.

7.2 The Gateway Guardian

All incoming data passes through a lightweight Gateway Guardian model (1B–3B parameters) that performs noise filtering, deduplication, cryptographic checksum verification (SHA-256), and taxonomy auto-tagging before engaging heavy domain specialists.

CHAPTER 8: SOVEREIGN ONCOLOGY VAULT & DEFENSE CASE STUDIES

8.1 Pattern #24: Oncology Combination Therapy Synergism

Investigating multidrug chemotherapy resistance in metastatic cancer, Sentinari identified dual-pathway synergy between statin HMG-CoA / SREBP-2 downregulation and cytotoxic agents. Mathematical validation under the Chou-Talalay Median-Effect Equation proved:

- Combination Index: **CI = 0.38 at ED_90** (Strong Synergism, threshold < 1.0)
- Dose Reduction Index: **DRI = 2.63-fold** (>62% reduction in toxic chemotherapy dosage)
- Day-30 Residual Tumor Volume: **148.2 mm³ vs. 1,842.5 mm³ control** (91.9% volumetric suppression).

8.2 Pattern #616 & #644: Critical Infrastructure & Defense

- **Pattern #616 (Power Grid Handover):** Dynamic Lyapunov state estimation reduced ping-pong oscillations by 48.4%, preventing blackout cascades.
- **Pattern #644 (Cognitive Radar Anti-Jamming):** HCAM pseudorandom frequency hopping achieved 94.85% packet delivery under 1,000 W broadband noise jamming.

Copyright © 2026 Dr. Pierre C. Wepener • University of Johannesburg • All Rights Reserved

CHAPTER 9: TROUBLESHOOTING & MASTER API DIRECTORY

9.1 System Launchers & Quickstart

• Double-click Sentinari-server\launch-all.bat to launch both the inference server (:8100) and the web core (:8099).
• Double-click launch-sentinari.bat to launch the web core independently.
• Health check: Navigate to http://127.0.0.1:8099/health.

9.2 Master REST API Route Directory

Endpoint Route	Method	Description & Return Payload
/health	GET	System status, active database dialect, and inference state.
/api/patterns/list	GET	JSON array of all discovered patterns and concertina metadata.
/api/patterns/rd-paper/{id}/report	GET	Full HTML 15-indicator Unified R&D; Validation Paper.
/api/patterns/assessment-matrix/view	GET	Interactive Chart.js 7-Dimensional Assessment Matrix with sliders.
/api/patterns/{id}/theoretical-engineering/download-prototype	GET	Download runnable Python prototype simulation script (.py).
/api/patterns/{id}/theoretical-engineering/download-matlab	GET	Download syntactically verified MATLAB numerical model (.m).
/api/qualia/vector-dynamics	POST	Calculate semantic vector trajectory, attractor S pull, and SI.

Copyright © 2026 Dr. Pierre C. Wepener • University of Johannesburg • All Rights Reserved

CHAPTER 10: MASTER ALPHABETICAL HELP INDEX & GLOSSARY

A comprehensive alphabetical cross-reference of features, mathematical terms, and system modules:

Index Term	Definition, Function & System Location
Algorithmic Distortion	The degradation of social and economic truth caused by engagement-optimizing algorithms. See Ch. 1 & White Paper Page 02.
Aristotelian Causality	The 4-cause epistemic framework (Material, Formal, Efficient, Final) anchoring Sentinari in reality. See Ch. 1.
Assessment Matrix (7-D)	The multi-criteria Gaussian mathematical framework evaluating K, I, D, H, S, E, and R. See Ch. 6.
Canvas Drawing Toolbar	Interactive annotation bar on the video tour canvas with freehand pen, bounding box, and auto-pause. See Ch. 3.
Chou-Talalay Synergism	Pharmacological median-effect equation proving combination drug efficacy (Pattern #24 CI=0.38). See Ch. 8.
Concertina Epistemic Cards	Expandable cards on the Pattern Discovery workspace showing field badges, confidence, and R&D; papers. See Ch. 4.
Declaration of Inadequacy	Article IV constitutional mandate: system states silence when empirical data is insufficient. See Ch. 1.
Dose Reduction Index (DRI)	Measure of how many-fold toxic drug doses can be lowered while maintaining therapeutic efficacy. See Ch. 8.
DRM Anti-Scraping Shield	Browser and print security interlocks disabling code copying, right-click, and unwatermarked printing. See Ch. 5.
Gateway Guardian	High-velocity 1B-3B model filtering noise, verifying hashes, and routing data before specialists. See Ch. 7.
Gaussian Bell Curve	The mathematical potency function $G_i(x_i) = A_i \cdot \exp(-(x_i - \mu_i)^2 / 2\sigma_i^2)$. See Ch. 6.
HDBSCAN Clustering	Hierarchical density clustering algorithm detecting embryonic patterns in superposition space. See Ch. 4.
Human Agency (H)	The supreme ethical anchor dimension (mu=7.8-9.0) protecting worker autonomy and labor dignity. See Ch. 1 & 6.
MATLAB Simulation (.m)	Verified numerical simulation scripts generated dynamically for every discovered pattern. See Ch. 5.
Net Sustainable Impact (NSIS)	The weighted aggregate score $NSIS = \Sigma W_i \cdot S_i(x_i)$ measuring total systemic health. See Ch. 6.
ONNX Inference Server	Local high-performance neural inference server running on port 8100 for offline rig-day execution. See Ch. 2.
OpenAlex API	Deterministic streaming integration connecting Sentinari to over 250 million academic publications. See Ch. 7.
pgvector 0.8.5	PostgreSQL extension enabling HNSW high-density vector similarity search in under 45 ms. See Ch. 2.
Pseudo-Dynamics Lab	Vector field laboratory modeling semantic inference as non-linear trajectory physics. See Ch. 2.
PubMed API	Primary biomedical API streaming peer-reviewed clinical research and oncology literature. See Ch. 7.
Raised, Not Scraped	The constitutional data ingestion doctrine prohibiting indiscriminate public web scraping. See Ch. 7.

Copyright © 2026 Dr. Pierre C. Wepener • University of Johannesburg • All Rights Reserved

Sliding Scale Rules	Coupled interdependence equations (Rule A, Rule B, Rule C) turning the Matrix into a flight simulator. See Ch. 6.
Stability Index (SI)	Metric evaluating semantic trajectory invariance toward the empirical attractor (SI >= 0.88). See Ch. 2.
Superposition Space	Intermediate latent vector space where cross-disciplinary affinities form embryonic patterns. See Ch. 4.
Theoretical Workbench	Dual simulator cockpit providing side-by-side Python prototype and MATLAB models in R&D; papers. See Ch. 5.
Victus Autonomous Engine	The internal challenger and forensic governor monitoring epistemic drift and state stability. See Ch. 2.
Volatility Anchors	Narrow-variance dimensions (Human Agency & Risk) where small deviations cause sharp NSIS drops. See Ch. 6.
Working Person's Shield	Constitutional mechanism providing workers with 24-36 month advance technological transition radars. See Ch. 1.
Yellow Checkmark	Animated glowing visual verification icon (particle pulse) certifying engineering milestone sign-off. See Ch. 3.
Zero-Cloud Doctrine	Architectural mandate requiring 100% of inference and proprietary research to run locally. See Ch. 1.

Copyright © 2026 Dr. Pierre C. Wepener • University of Johannesburg • All Rights Reserved